

NinaivX — Literature Gap Analysis

What prior work lacks, the gaps it reveals, and the requirements those gaps generate · Ashwin Kumar Ramesh Kumar · MSc AI, University of Sheffield

This document distils the Literature Survey into a gap analysis and links it directly to the Requirements Specification. Each gap carries an ID (**G1–G9**) used identically in the **Requirements (MoSCoW)** document, and each row shows the **requirements** that gap generated. Reference numbers [n] match the Literature Survey. Where a commercial system does not publicly document a safeguard, this is stated as “not documented” rather than asserted as absent.

A. System-by-system — what it lacks, and what NinaivX does about it

Prior system / work	What it offers	Gap — what it lacks	Gap ID	How NinaivX addresses it	Req.	Ref
HereAfter AI	An interactive “life story” answering in the deceased's own recorded voice.	Replays pre-recorded, consented clips only — cannot answer questions it was never asked.	G1	Generative, real-time conversation that responds to novel questions, with memory across sessions.	M5, M6, M9	[16]
Project December	A GPT-3 “griefbot”; free-form chat built from a person's messages.	Text-only; unconstrained model can fabricate memories; a single anecdotal case with no evaluation.	G2, G3, G8	Full voice loop; air-gapped and grounded to curb fabrication; safety design empirically evaluated.	M7–M9, M10, M13, M15	[17]
Replika	A popular AI companion; text chat; emotional relationship.	Text-heavy; engagement-driven design linked to emotional dependence and harm; not designed for grief.	G2, G5	Voice-first interaction; oriented to comfort and closure, with no engagement-maximising mechanics.	M7–M9, S10	[12–15]
Xiaolce	An empathetic social chatbot engineered for long-term engagement.	Text-based; optimised to maximise engagement; no bereavement safeguards.	G2, G5	Voice; closure-oriented design rather than engagement maximisation.	M7–M9, S10	[11]
Microsoft patent	A method to build a chatbot of a specific person (incl. deceased) from their data.	A design claim, not a deployed or reproducible system; consent and privacy for the deceased unresolved.	G6, G7	A working, openly documented implementation with authentication and strict per-user data isolation.	M1, M11, M14	[18]
You, Only Virtual (YOV)	Commercial voice griefbot; phone-call interface, voice cloning, memory, generative conversation.	Frames grief as “a problem to be solved”; no documented safety or transparency; closed/proprietary; unevaluated; deceased-only.	G3, G4, G5, G7, G8, G9	Air-gap safety; AI disclosure; closure-oriented stance; open architecture; empirical evaluation; adds a companion mode.	M10, M14, M15, S9, S10, M3, S1	[26–27]
Unconstrained LLMs (e.g. GPT-	Highly fluent, human-like text generation.	Prone to “hallucination” — inventing facts or opinions the real	G3	Deceased persona air-gapped from the internet (code-level)	M4, M10, M13	[3][19]

3)		person never had.		and grounded in stored personality + memory.	
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B. Gap register (G1–G9) — the requirements each gap generated

ID	Gap in existing work	Why it matters	Requirements generated	Status	Ref
G1	Static memorials cannot hold a real conversation — they only replay recordings.	The bereaved cannot ask anything new; the experience breaks down.	M5, M6, M9, S5 (C2 future)	✓ Delivered	[16]
G2	Existing griefbots/companions are text-only, lacking natural voice interaction.	Voice, not text, carries the presence of a lost person.	M7, M8, M9, S2, S3, S8 (C2 future)	✓ Delivered	[17][12–15][11]
G3	Unconstrained generative models fabricate things the real person never said.	An AI imitating a real dead person must not invent memories.	M4, M10, M13 (C1 future)	✓ air-gap · ● grounding	[3][19][17][26]
G4	No AI-transparency disclosure documented, although the ethics literature requires it.	Grieving users deserve to know they are speaking with an AI.	S9	✓ Delivered	[25][26]
G5	Engagement-driven design fosters dependency; systems are not designed for grief and closure.	Maximising engagement can block healthy mourning.	S10 (assessed via M15)	● By design; evaluation pending	[11][15][26]
G6	Sensitive data of the deceased is handled opaquely; consent and privacy unresolved.	A dead person's voice and data are deeply sensitive and cannot consent.	M1, M10, M11, C5	✓ Delivered	[18][23][24]
G7	Systems are closed and proprietary; the architecture cannot be inspected or reproduced.	Nothing can be verified, criticised or built upon.	M14	✓ Delivered	[18][26]
G8	No published evaluation of safety or effectiveness.	Products reach grieving users with no evidence they are safe.	M15	⌘ Planned (Aug evaluation)	[17][26]
G9	Scope is limited to the deceased; no companion mode addressing loneliness.	Loneliness is a distinct, widespread need the field ignores.	M3, M4, S1	✓ Delivered	[26]

Status key: ✓ implemented · ● partial/basic · ⌘ planned. Gap IDs (G1–G9) and requirement IDs (M/S/C) are shared with the Requirements (MoSCoW) document, which contains the full specification and traceability matrix. References [n] correspond to the Literature Survey (verified June 2026). Commercial systems' undocumented safeguards are reported as “not documented”, not asserted as missing.